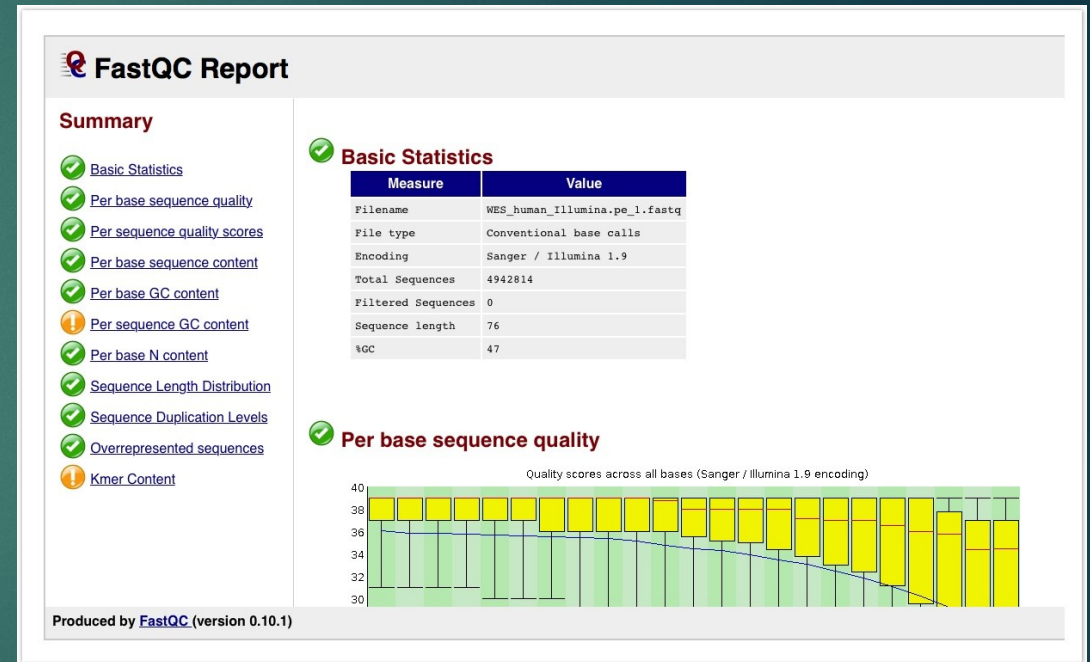


ChIP-Seq Best Practices

JAMES ROUSE

Best practices during quality control of sequence reads

- ▶ **Purpose:** Assess the quality of the reads and ensure that the sequencing data is of high quality and reliable for downstream analysis.
- ▶ **FastQC:** Identifies potential issues such as poor read quality, adapter contamination, and sequence biases.
- ▶ **MultiQC:** Aggregates QC reports to identify trends, batch effects, and outliers, providing a summary across all samples
- ▶ **Fastq_screen:** Aligns reads to multiple reference genomes/databases. Good way to detect contamination.
- ▶ Resequencing often recommended if the results of the FastQC report are poor.



Best practices during adapter trimming

- ▶ **Adapter trimming:** Adapter trimming is highly recommended for ChIP-seq, since it deals with shorter DNA sequences. Trimmomatic and CutAdapt are two common tools used for trimming.
- ▶ Not trimming the adapters can result in decreased mapping quality and produce artificial peaks

Best practices during alignment

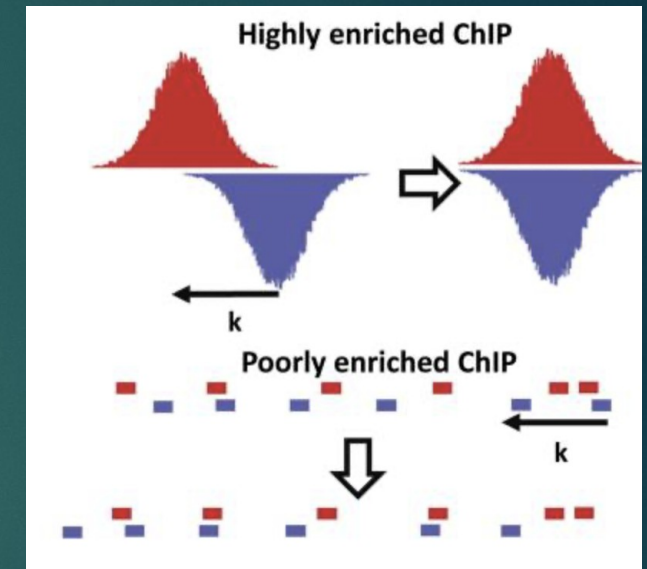
- ▶ **BWA** and **Bowtie** as main tools for ChIP-Seq: These tools are useful for mapping short reads with high accuracy and stringent control over mismatches.
- ▶ [Zhang et al](#): **Bowtie** often preferable over BWA for ChIP-Seq since Bowtie only reports uniquely mapped reads, resulting in higher alignment accuracy compared to BWA.
- ▶ Other alignment tools like STAR's advantage of being able to detect novel splice junction and sequencing long reads is less important in ChIP-Seq since it sequences short DNA rather than RNA.
- ▶ Removal of low quality reads and reads with a low mapping score is recommended.

ChIP-seq quality control of aligned reads

- ▶ Quality control after alignment mainly consists of removing duplicates and assessing signal versus noise
- ▶ **Removing Duplicates**
 - ▶ **Picard** marks PCR duplicates to remove non-informative reads, ensuring only unique fragments are analyzed.
 - ▶ **NRF** = distinct uniquely mapped reads / total uniquely mapped reads.
 - ▶ High NRF (>0.8) indicates good library complexity, while a low NRF (<0.5) suggests high duplication and low quality, impacting reliability.
- ▶ **Filter uninformative reads:** Filtering out low quality reads and reads that map to the mitochondrial genome leaves us with a set of more informative reads.

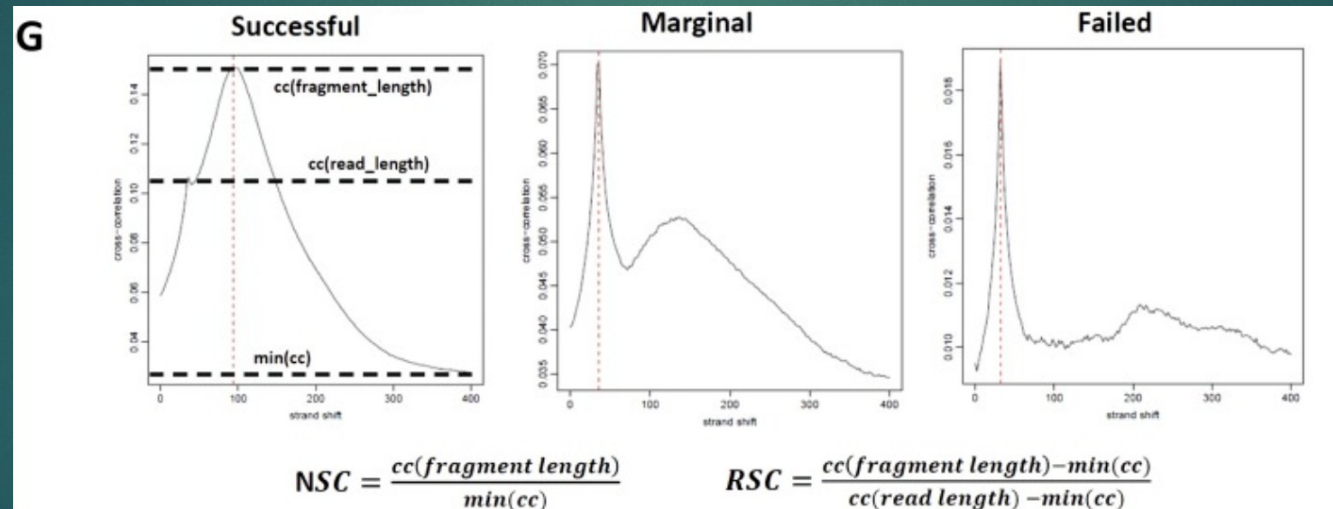
Quality assessment after alignment

- ▶ **Cross-Correlation Analysis in ChIP-seq**
 - ▶ Cross-correlation is used to assess **signal enrichment** by evaluating the correlation between forward and reverse reads.
 - ▶ It helps estimate **fragment length**, indicating successful DNA shearing and enrichment quality.
 - ▶ Typically performed after alignment, it ensures data quality before peak calling, using metrics like NSC and RSC. Tools like **PhantomPeakQualTools** can be used to calculate these metrics



ChIP-seq quality assessment

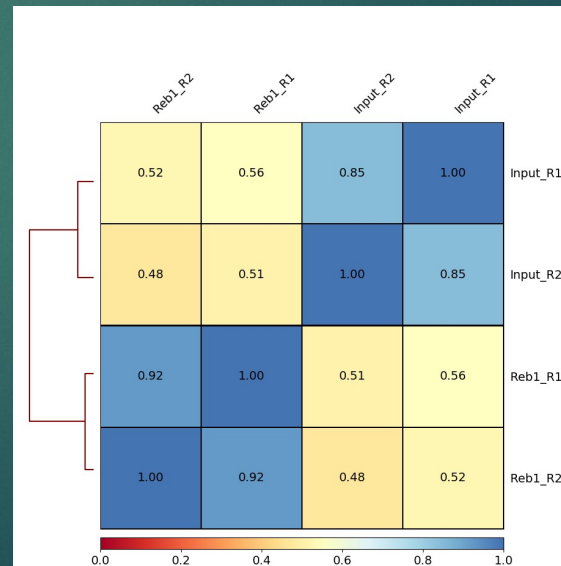
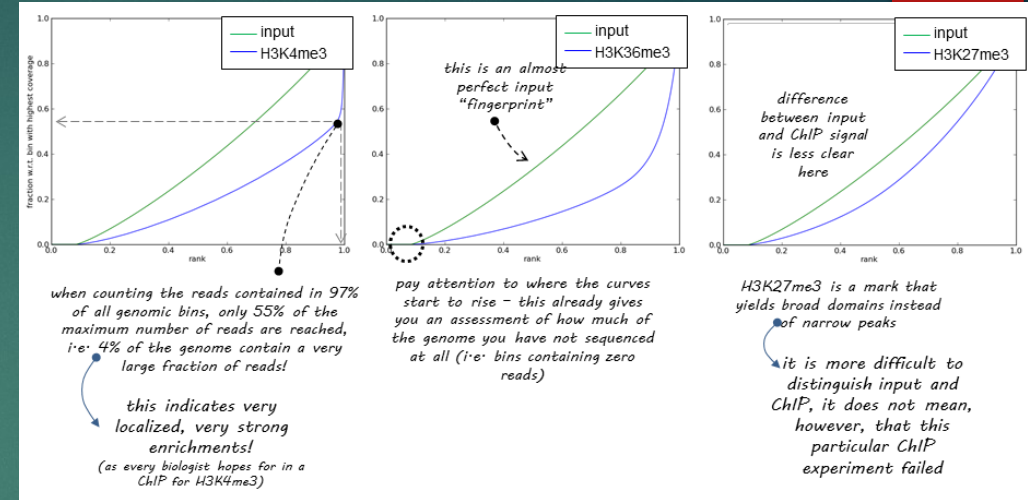
- ▶ Use PhantomPeakQualTools (SPP) for NSC and RSC metrics: NSC > 1.05 and RSC > 0.8: Good quality. RSC < 0.5: Poor quality.
- ▶ For experiments with NSC values below 1.05 and RSC below 0.8, ENCODE recommends adding another replicate.



[Landt et al](#)

ChIP-seq quality assessment

- ▶ Use **DeepTools** to plot **fingerprint plot** of ChIP-seq data and input data. DeepTools can also be used to check **correlation** between samples.
- ▶ Steep rise towards the right side of the graph represents high number of reads in a small portion of the genome. (Strong enrichment)
- ▶ High correlation between treatments and control imply strong signal.



Best practices during peak calling

▶ How does peak calling work?

- ▶ Aligned reads are analyzed to identify regions with significantly higher coverage compared to the background.
- ▶ These regions are called **peaks**, which indicate potential binding sites or areas of interest.
- ▶ Requires both the **ChIP sample** (target protein) and a **control sample** (e.g., input DNA) to accurately differentiate between true signal and background noise.

Best practices during peak calling

- ▶ [Jeon et al.](#) - **CisGenome**, **MACS1**, **MACS2**, **PeakSeq**, and **SISSRs** were compared on 12 histone modifications in human embryonic stem cells, covering **narrow**, **broad**, and **mixed-source** histone modifications.
- ▶ **Broad modifications** (e.g., H3K27me3) required more reads to achieve saturation, with **MACS2 (broad option)** covering larger loci.
- ▶ **MACS1** showed better reproducibility across biological replicates, particularly for narrow marks like **H3K4me3** and **H3K9ac**.
- ▶ **PeakSeq** and **CisGenome** are reliable for consistency and robustness, even in high-noise conditions.
- ▶ **MACS2** was highly sensitive to noise, with its ability to recapture true peaks significantly reduced, especially for histone marks with inherently weak signals such as H3K4ac, H3K56ac, and H3K79me1

Best practices during peak calling

- ▶ [Thomas et al.](#) – Investigated specific features that define the best peak calling algorithms
- ▶ **Best Performers: BCP and MACS2** excelled in transcription factor binding data; **GEM** performed best for identifying peaks near binding motifs; **BCP and MUSIC** were best for histone marks.
- ▶ **Multiple window sizes** for scanning the genome enhanced peak detection.
- ▶ The **Poisson test** was more effective for ranking peaks compared to the Binomial test.
- ▶ **Input Control: Input control** is still essential for filtering out false positives during peak ranking, but only using ChIP signal improved power.

QC after Peak calling

- ▶ **Check Gene Promoter Enrichment:** Confirm protein enrichment at known gene promoters and enhancers to validate ChIP-seq data quality.
- ▶ **FRiP (Fraction of Reads in Peaks):** Calculate the fraction of reads within called peaks. Higher FRiP scores indicate better enrichment for transcription factors. Tool: ChIPQC
- ▶ **IDR (Irreproducible Discovery Rate):** Assess and retain reproducible peaks across replicates to ensure data reliability. Tool: IDR

ChIPQC

- ▶ **ChIPQC** can create visualizations and measures for several quality metrics, including:
 - ▶ Reads: Total number of reads in the sample.
 - ▶ Dup%: Percentage of duplicate reads.
 - ▶ ReadL: Read length.
 - ▶ FragL: Fragment length.
 - ▶ RelCC: Relative cross-correlation, assessing enrichment quality.
 - ▶ SSD: Strand shift distance, related to enrichment signal.
 - ▶ FRiP: Fraction of reads in peaks, indicating signal enrichment.
 - ▶ RiBL%: Percentage of reads overlapping ENCODE blacklist regions.

ID	Tissue	Factor	Condition	Replicate	Reads	Dup%	ReadL	FragL	RelCC	SSD	RiP%	RiBL%
Nanog.Rep1		Nanog		1	969186	0	36	121	1.4	2	4.1	2.9
Nanog.Rep2		Nanog		2	2283248	0	36	136	2.7	1.9	6.5	1.5
Pou5f1.Rep1		Pou5f1		1	1085316	0	36	164	2.3	2.6	3.9	3.2
Pou5f1.Rep2		Pou5f1		2	1995385	0	36	151	5.8	3	0.92	2.6
Nanog-Input1	NA	NA	NA	NA	4080970	0	36	73	0.2	4.3	NA	5.2
Nanog-Input2	NA	NA	NA	NA	1817134	0	50	104	0.64	0.91	NA	0.56

MACS2 – Zhang et al

- ▶ Estimates fragment length and shifts the reads
- ▶ The reads from the input (control) and ChIP sample are normalized to account for sequencing depth.
- ▶ Lambda, which estimates background noise is estimated over windows of different sizes in the background and takes the max, most conservative to reduce the risks of false positive.
- ▶ Lambda is estimated locally, which is a strength of MACS.
- ▶ Sliding window across the genome to identify regions where the read density significantly exceeds the background level.
- ▶ Poisson distribution models background noise to identify genomic regions with significantly more reads than expected by chance

Peak annotation and motif analysis

- ▶ **HOMER** or **ChIPseeker** to map peaks to nearby genes or genomic features (promoters, exons, introns) and determine their biological relevance.
- ▶ **HOMER Motif Analysis**: HOMER identifies DNA motifs enriched in ChIP-seq peak regions, helping to reveal binding patterns of transcription factors through de novo discovery and known motif searches.

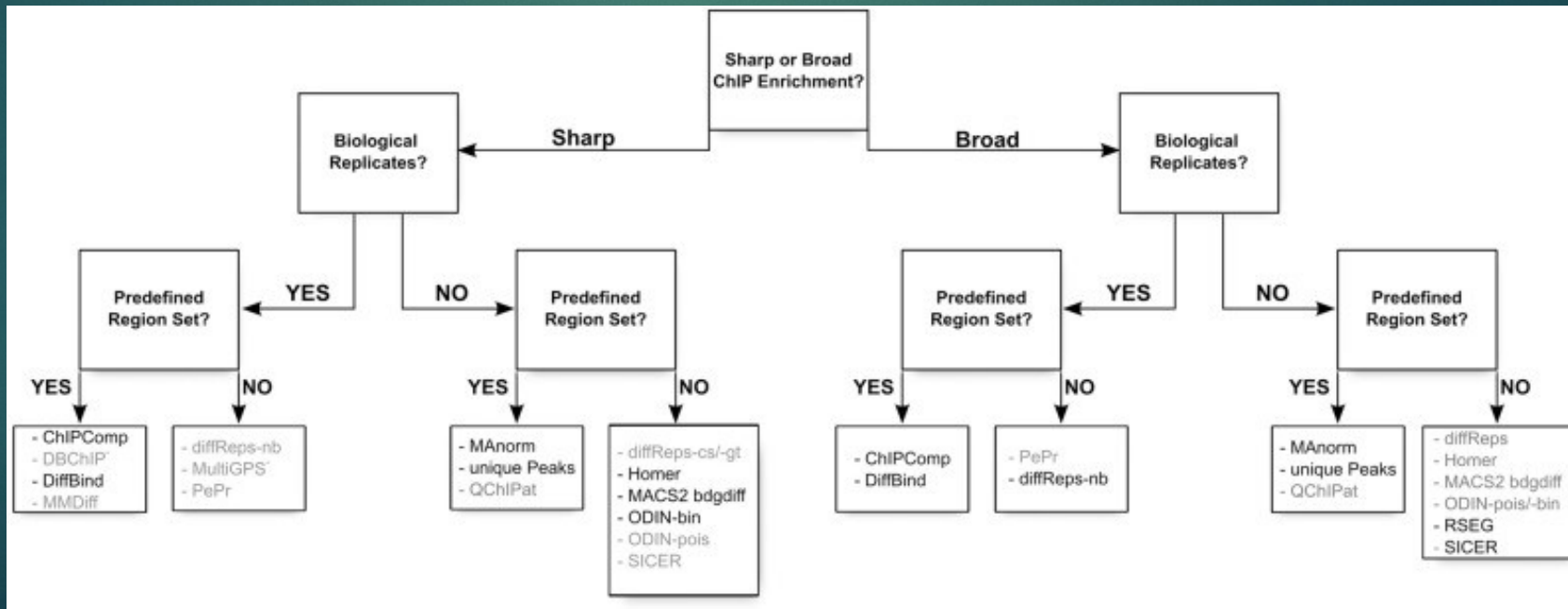
Homer de novo Motif Results

[Known Motif Enrichment Results](#)
[Gene Ontology Enrichment Results](#)
 If Homer is having trouble matching a motif to a known motif, try copy/pasting the matrix file into [STAMP](#)
 More information on motif finding results: [HOMER](#) | [Description of Results](#) | [Tips](#)
 Total target sequences = 15312
 Total background sequences = 34047
 * - possible false positive

Rank	Motif	P-value	log P-value	% of Targets	% of Background	STD(Bg STD)	Best Match/Details	Motif File
1		1e-1835	-4.228e+03	28.11%	5.16%	37.7bp (63.1bp)	NFKB-p65(RHD)/GM12787-p65-ChIP-Seq/Homer More Information Similar Motifs Found	motif file (matrix)
2		1e-1716	-3.953e+03	34.50%	8.65%	47.8bp (62.6bp)	PB0058.1_Spt1.1 More Information Similar Motifs Found	motif file (matrix)
3		1e-1585	-3.651e+03	28.85%	6.39%	41.8bp (62.8bp)	MA0102.1_Cebpa More Information Similar Motifs Found	motif file (matrix)
4		1e-1004	-2.314e+03	25.07%	7.22%	49.2bp (61.0bp)	NF-E2/ZIP/K562-NFE2-ChIP-Seq/Homer More Information Similar Motifs Found	motif file (matrix)
5		1e-262	-6.039e+02	5.52%	1.27%	47.8bp (59.7bp)	Oct2(POU/Homeobox)/Bcell-Oct2-ChIP-Seq/Homer More Information Similar Motifs Found	motif file (matrix)
6		1e-198	-4.565e+02	6.42%	2.09%	49.9bp (53.2bp)	c-Jun-CRE(bZIP)/K562-cJun-ChIP-Seq/Homer More Information Similar Motifs Found	motif file (matrix)
7		1e-146	-3.370e+02	7.66%	3.30%	51.7bp (58.9bp)	RUNX1(Runt)/Jurkat-RUNX1-ChIP-Seq/Homer More Information Similar Motifs Found	motif file (matrix)

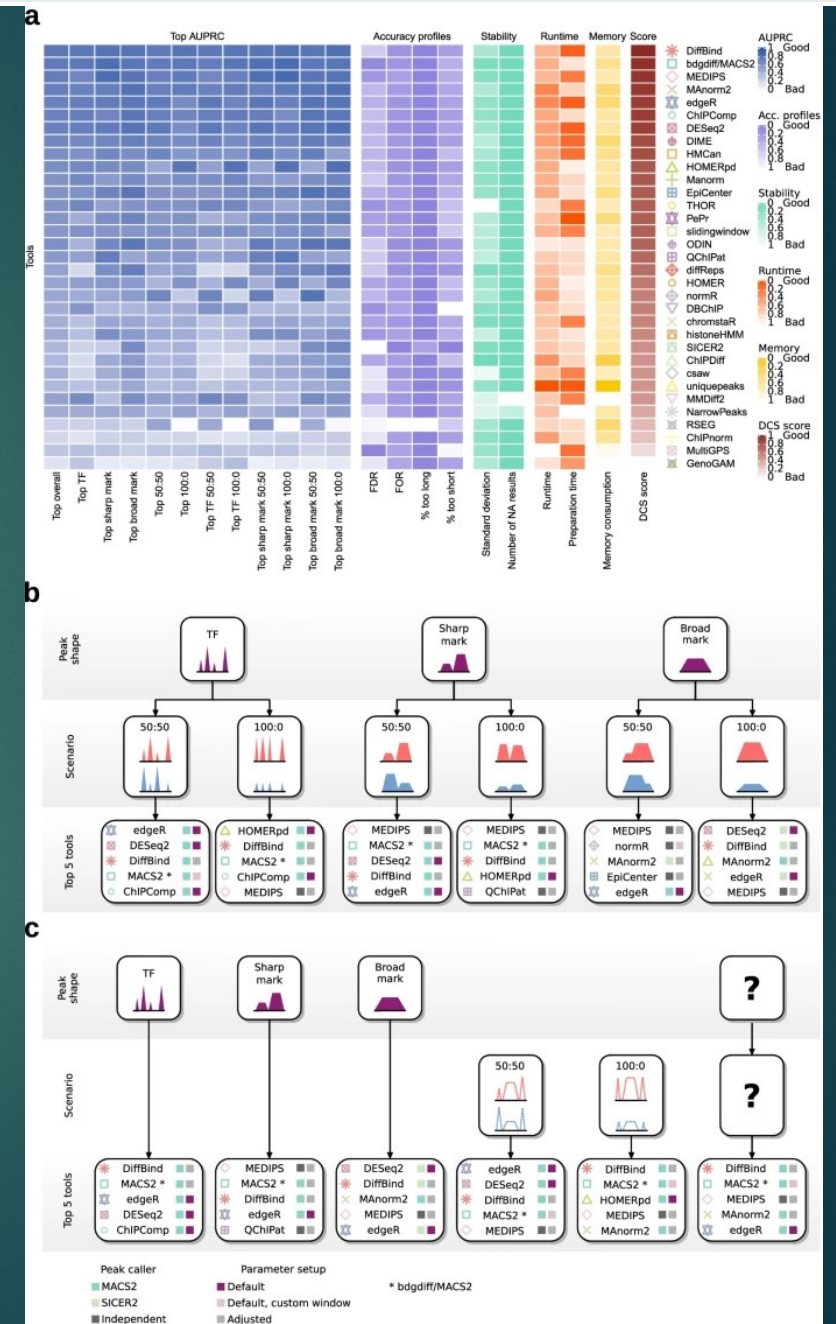
Differential Peaks

- ▶ [Steinhauser et al](#) - Compared the effectiveness of 13 different differential peak tools.
- ▶ The tools had varying levels of performance depending on the data set:
 - ▶ Shape of the signal (sharp peaks or broad enrichments)
 - ▶ Presence of replicates
 - ▶ Presence of an external set of regions of interest



Differential Peaks

- ▶ [Eder et al.](#) - The performance of differential ChIP-seq tools varies significantly depending on peak type and biological scenario.
- ▶ DiffBind, bdgdiff (MACS2), and ChIPComp consistently performed well across multiple conditions.
- ▶ Different tools perform significantly better or worse depending on whether they are used for transcription factors (TFs), broad histone marks, or sharp marks

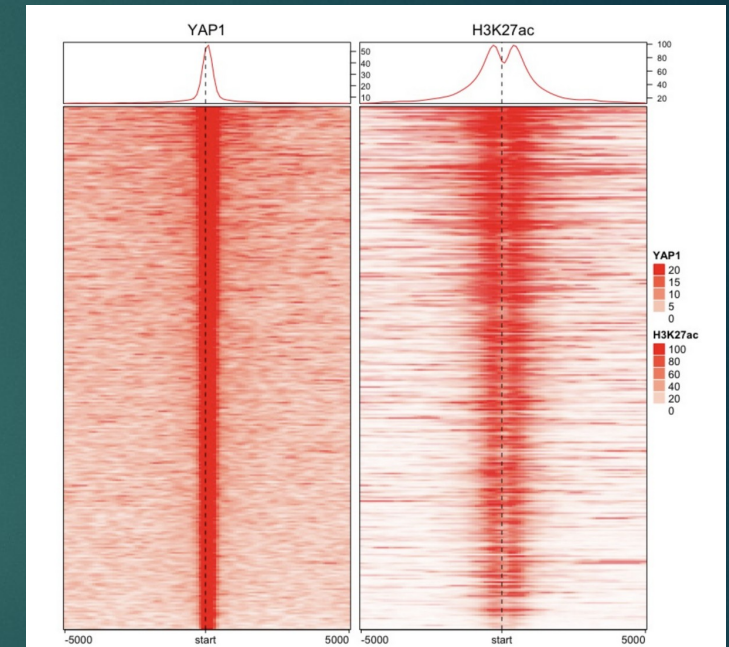
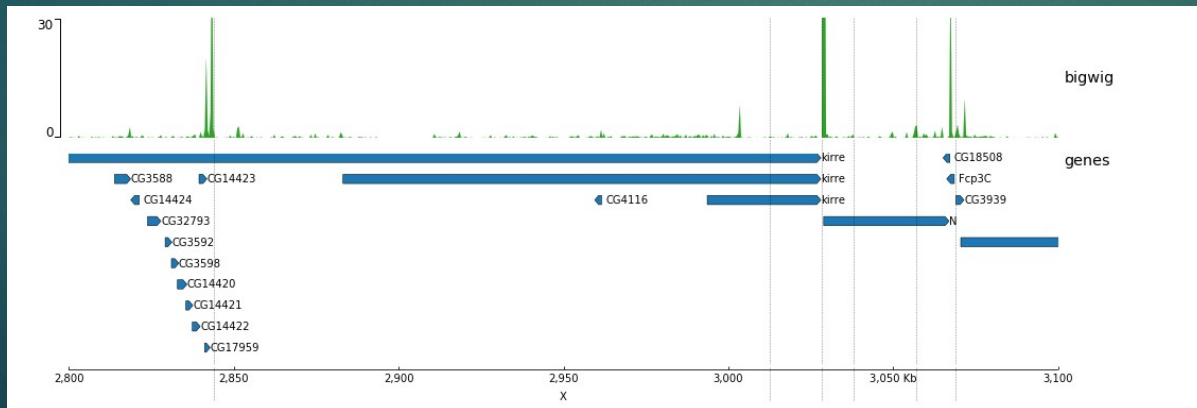


Factors to consider when picking a DCS tool

- ▶ Peak Type and Data Scenario: The type of peaks (e.g., TFs, sharp, broad) and the regulation scenario (e.g., equal vs. global changes) should guide tool selection, as different tools are optimized for specific peak characteristics and scenarios.
- ▶ Based on both studies, DiffBind, MACS2 (bdgdiff), ChIPComp, and HOMER generally performed the best across a variety of experimental scenarios and biological conditions.

Visualization

- ▶ **Signal Tracks & Heatmaps:** **Signal tracks** visualize read coverage across genomic regions, and **heatmaps** help compare enrichment profiles across loci.
- ▶ **pyGenomeTracks** allows multi-layered visualization of **ChIP-seq signals** across the genome.
- ▶ IGV (Integrative Genomics Viewer) is a flexible and widely used tool for visualizing ChIP-Seq data



Summary

- ▶ Alignment Tools: Bowtie and BWA are primary tools for aligning short reads. Bowtie is often preferred for its focus on uniquely mapped reads, improving accuracy.
- ▶ Peak Calling: MACS2 is the best general tool for identifying significant regions (peaks). For transcription factors, MACS2 and GEM perform well, while BCP and MUSIC are recommended for histone marks.
- ▶ QC Tools: FastQC for initial quality check, MultiQC for aggregating reports, and PhantomPeakQualTools for calculating NSC and RSC post-alignment.
- ▶ Differential Peak Tools: Tools like DiffBind, MACS2 (bdgdiff), HOMER, and ChIPComp are highly effective for identifying differential peaks. Factors like sharp or broad enrichment or global vs. equal changes can impact which tool is best to use.